

Uniform complex border-rank bounds from all-cut Schmidt ranks

Abstract

Every Schmidt rank is a lower bound on CP border rank. We prove a converse uniform in tensor order: if all Schmidt ranks of a complex tensor are at most r , its CP border rank is at most $\exp(O(r^2 \log r))$, independently of the number and dimensions of its factors. This answers the border-rank question of Florido-Llinàs et al. (PRX Quantum 2025) for matrix product states under permutations. Product contractions retain an invertible core of a maximal-rank flattening on boundedly many factors. The other cut constraints force a common matrix-product representation, with noncommutativity confined to boundedly many factors. The remaining factors multiply in a finite-dimensional commutative algebra. Decomposing this algebra into local components, we use nilpotence of their maximal ideals and polarization to express the products through bounded-degree coefficients of product curves. Finite differences give border-rank degenerations of size independent of the product length.

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1 Introduction

A decomposition into b product tensors gives a matrix of rank at most b across every bipartition. The same bound holds for limits of such decompositions, because matrix rank conditions are closed. Thus Schmidt ranks provide necessary lower bounds on CP border rank. The converse is less clear: low-rank decompositions across different cuts need not use compatible product factors. We ask whether bounding *all* Schmidt ranks is nevertheless enough to bound border rank when the number of parties is allowed to grow.

This question arises naturally for matrix product states (MPS). A bound on the Schmidt ranks of prefix cuts gives an open-boundary MPS in one chosen ordering. Requiring the same bound in every ordering imposes it on every bipartition. Florido-Llinàs et al. study this class of states and obtain product structure under further assumptions on their MPS representations [1]. They ask whether exact MPS under permutations can have border rank growing without bound with the system size [1, Outlook]. We show that they cannot, without imposing injectivity, translation invariance, or permutation invariance of the state.

Exact tensor rank would give a false converse even at Schmidt rank two. For linearly independent e_0, e_1 , the N -party W tensor, $N \geq 2$, has tensor rank N and every Schmidt rank at most two [1, Sections I.C and VIII]. Yet

$$W_N = \lim_{t \rightarrow 0} \frac{(e_0 + te_1)^{\otimes N} - e_0^{\otimes N}}{t}$$

has border rank two. The distinction is that all N possible positions of e_1 are obtained from one coefficient of a product curve. Our proof extends this coefficient-extraction mechanism to the algebraic structure forced by the all-cut constraints.

The usual local support reduction does not give the required uniformity. Singleton cuts of rank at most r allow every local space to be replaced by one of dimension at most r , but expanding the

resulting tensor still gives only r^{N-1} product terms. More general exact-rank bounds from separating collections of cuts also depend on N [6]. Nor does bounded-degree definability of border-rank loci resolve the issue. Draisma and Kuttler show that, for each fixed border-rank threshold, equations of bounded degree suffice uniformly in the tensor spaces [5]. Here the threshold itself must be deduced from flattening minors alone.

We obtain that threshold by first finding a small set of factors that preserves the full information of one maximal-rank cut. Product contractions can retain its two r -dimensional Schmidt supports while leaving at most $r - 1$ factors on either side. The contracted tensor is an invertible matrix. Keeping this matrix visible inside each further flattening that separates the two anchor sets forces its other blocks to factor through it. Consequently, the remaining local variables admit matrix representatives whose products can be taken in any order. Although the matrices attached to one factor may fail to commute internally, each such factor strictly reduces a common centralizer in the r^2 -dimensional matrix space. Only boundedly many factors can do so.

Expanding these exceptional factors leaves arbitrarily long products in a commutative algebra of bounded dimension. Diagonalizing the algebra would turn these products into sums of product tensors, but commutative algebras with nonzero nilpotent elements need not be diagonalizable. Instead, we decompose the algebra into local factors and use nilpotence to bound the number of non-scalar terms that survive in a product. Polarization collects terms of the same degree into coefficients of product curves, just as in the W_N example. The number of evaluations needed to recover such a coefficient depends on its degree, rather than on N . This yields the uniform border-rank bound below.

1.1 Main theorem

Throughout, all vector spaces are finite-dimensional over $\mathbb{C}$. Write $[N] = \{1, \dots, N\}$, put $V_I = \bigotimes_{i \in I} V_i$, and set $V_\emptyset = \mathbb{C}$. We identify tensor products that differ only by a reordering of their factors. For $S \subseteq [N]$, the flattening of $T \in V_{[N]}$ across $S \mid S^c$ is

$$\text{Flat}_{S|S^c}(T) : V_S^* \longrightarrow V_{S^c}, \quad \xi \longmapsto (\xi \otimes \text{id})(T).$$

Its matrix rank is the Schmidt rank across that cut. The CP rank $\text{R}(T)$ is the least number of decomposable tensors whose sum is T . The complex CP border rank $\underline{\text{R}}(T)$ is the least integer $b \geq 0$ such that T belongs to the Zariski closure of the tensors of CP rank at most b . Over $\mathbb{C}$, this is also their Euclidean closure. In particular, $\underline{\text{R}}(T) \leq b$ exactly when T is a limit of tensors of rank at most b . We set $\text{R}(0) = \underline{\text{R}}(0) = 0$.

For an integer $D \geq 1$, the hypothesis of interest is

$$\text{rank Flat}_{S|S^c}(T) \leq D \quad (\emptyset \neq S \subsetneq [N]). \quad (1)$$

For a fixed ordering of the factors, the minimal bond dimensions of an open-boundary matrix product state (MPS) are the Schmidt ranks across its prefix cuts. Indeed, a bond gives a factorization of the corresponding flattening, while successive Schmidt decompositions attain these ranks. Every bipartition is a prefix cut in some ordering. Thus (1) is equivalent to admitting an open-boundary MPS of bond dimension at most D in every ordering.¹ No bound on the ambient local dimensions is needed: the singleton cuts already bound the dimensions of the local supports.

Set $F(1) = 1$, and for integers $q \geq 2$ define

$$F(q) := \sum_{k=0}^{q-1} (k+1) \binom{q+k-2}{k} = \binom{2q-2}{q-1} + (q-1) \binom{2q-2}{q-2}. \quad (2)$$

¹This numerical equivalence uses open boundaries. A periodic MPS with bond dimension at most D gives Schmidt rank at most D^2 across a contiguous bipartition, since that cut crosses two bonds.

The equality and the monotonicity properties used below are proved in Lemma 3.1.

Theorem 1.1 (Uniform border-rank bound). *Let $N \geq 2$, let $0 \neq T \in V_1 \otimes \cdots \otimes V_N$, and put*

$$r := \max_{\emptyset \neq S \subsetneq [N]} \text{rank Flat}_{S|S^c}(T).$$

If $r = 1$, then T is decomposable. If $r \geq 2$, then

$$\underline{R}(T) \leq r^{r^2+2r-3} F(r^2). \quad (3)$$

Let $B_*(D)$ be the supremum of $\underline{R}(T)$ over all nonzero complex tensors of order $N \geq 2$ satisfying (1), allowing both N and the local dimensions to vary. Theorem 1.1 shows that this supremum is finite. It therefore is the smallest integer bound valid uniformly over all such tensor spaces. Define $\widehat{B}(1) = 1$ and, for integers $D \geq 2$,

$$\widehat{B}(D) := D^{D^2+2D-3} \left[\binom{2D^2-2}{D^2-1} + (D^2-1) \binom{2D^2-2}{D^2-2} \right]. \quad (4)$$

Corollary 1.2 (Uniform secant containment). *For every integer $D \geq 1$, $B_*(D) \leq \widehat{B}(D)$. Equivalently, if*

$$X = \text{Seg}(\mathbb{P}V_1 \times \cdots \times \mathbb{P}V_N) \subseteq \mathbb{P}(V_{[N]}),$$

where $N \geq 2$ and the V_i are nonzero, then, set-theoretically,

$$\left\{ [T] : \text{rank Flat}_{S|S^c}(T) \leq D \text{ for every } \emptyset \neq S \subsetneq [N] \right\} \subseteq \sigma_{\widehat{B}(D)}(X).$$

Here $\sigma_b(X)$ is the Zariski closure of the union of the linear spans of at most b points of X .

Proof. For $2 \leq r \leq D$, both the base and the exponent of r^{r^2+2r-3} increase with r , as does $F(r^2)$. Theorem 1.1 therefore gives the claimed bound. The case $r = 1$ is included in the same theorem. The secant formulation follows from $[T] \in \sigma_b(X)$ if and only if $\underline{R}(T) \leq b$. $\square$

The rank-one case gives $B_*(1) = 1$, and the conclusion is already known with the sharp bound at $D = 2$. Raicu's theorem states that the ideal of the second secant of a Segre variety is generated by the 3×3 minors of its flattenings [2, Corollary 4.2]; hence $B_*(2) = 2$. The analogous equality with the matching secant fails at $D = 3$. Qi's description of the third secant uses Strassen's equations in addition to the 4×4 flattening minors [3, Theorem 1.4]. More concretely, a general $3 \times 3 \times 3$ tensor has border rank 5: the fourth secant of $\text{Seg}(\mathbb{P}^2 \times \mathbb{P}^2 \times \mathbb{P}^2)$ is defective, whereas its fifth secant fills the ambient space [4, Section 4]. Every flattening of such a tensor has rank at most 3. Thus

$$5 \leq B_*(3) \leq \widehat{B}(3) = 3^{12} F(9) = 55,477,125,990.$$

The estimate is consequently far from sharp already at $D = 3$. Its significance is that $B_*(D)$ is finite for every fixed D , even as the order and local dimensions vary. The proof occupies Sections 2–4. Proposition 3.2, which bounds arbitrary-length products in a commutative algebra, also gives a cactus-to-secant containment in Section 5. That section records the connection with any-order algebraic branching programs as well.

2 From all-cut constraints to a matrix-product structure

Singleton cuts first remove the dependence on local dimensions. For $0 \neq T \in V_{[N]}$, let

$$E_i := \text{im Flat}_{[N] \setminus \{i\} \{i\}}(T) \subseteq V_i.$$

Then $T \in E_1 \otimes \cdots \otimes E_N$. To see this, choose projections $p_i : V_i \rightarrow E_i$ restricting to the identity on E_i . The definition of E_i implies that applying p_i in the i -th factor leaves T unchanged, so applying all of them does as well. Inclusions and these projections show that restricting to the E_i preserves all flattening ranks and CP border rank: local linear maps cannot increase either quantity, and the maps in the two directions fix T . Under the all-cut bound, $\dim E_i \leq r$. We may therefore assume $\dim V_i \leq r$ whenever this is useful.

A contraction that preserves one nonzero tensor may collapse the other vectors in its Schmidt support. We therefore need injectivity on the whole support. The next lemma provides it after retaining at most $r - 1$ factors. The proof grows the retained set until the contraction has full rank: if adding any one factor failed to increase that rank, variation over a product basis would force all detected dual spaces to coincide, contradicting the dimension of the original support. This is the quantitative contraction step used here; related product-contraction arguments appear in [5, Lemma 4.4].

Lemma 2.1 (Faithful product contraction). *Let I be a finite index set and let $E \subseteq V_I$ have dimension $r \geq 1$. There are $J \subseteq I$, with $|J| \leq r - 1$, and covectors $b_i \in V_i^*$ for $i \in I \setminus J$ such that*

$$\left(\text{id}_{V_J} \otimes \bigotimes_{i \in I \setminus J} b_i \right) \Big|_E : E \longrightarrow V_J$$

is injective.

Proof. For $J \subseteq I$ and a tuple $\alpha = (\alpha_i)_{i \in I \setminus J}$ of complementary covectors, define

$$L_J(\alpha) := \left\{ \left(\beta \otimes \bigotimes_{i \in I \setminus J} \alpha_i \right) \Big|_E : \beta \in V_J^* \right\} \subseteq E^*, \quad d(J) := \max_{\alpha} \dim L_J(\alpha).$$

The integer $d(J)$ is the largest rank of a product contraction of E leaving the factors in J . It is attained on a nonempty Zariski-open set of complementary tuples, since rank is detected by matrix minors. Product covectors span V_I^* , so $d(\emptyset) = 1$ and $d(I) = r$. Moreover, $d(J \cup \{i\}) \geq d(J)$ whenever $i \notin J$.

Suppose that $d(J) = d < r$. We claim that $d(J \cup \{i\}) > d$ for some $i \notin J$. Otherwise every one of these integers equals d . For each $i \in I \setminus J$, choose a basis $\mathcal{B}_i$ of V_i^* so that

$$\dim L_J(\alpha) = d \quad \text{whenever} \quad \alpha \in \prod_{i \in I \setminus J} \mathcal{B}_i. \tag{5}$$

Such bases exist. For each fixed choice of one basis position in every factor, the maximal-rank condition is a nonempty Zariski-open condition on the product of the spaces of bases. It is nonempty because a maximal-rank tuple consists of nonzero covectors and each of those covectors can be extended to a basis. There are only finitely many choices of positions, and the product of the spaces of bases is irreducible, so all these open conditions can be satisfied together.

If two tuples in this finite product differ only in their i -th entry, both corresponding spaces L_J are contained in $L_{J \cup \{i\}}$ evaluated at their common remaining entries. That space has dimension

at most $d(J \cup \{i\}) = d$, while both subspaces have dimension d by (5). They are therefore equal. Changing one entry at a time shows that all spaces in (5) are a single subspace $L \subseteq E^*$. As β varies over V_J^* and α varies over this product of bases, the covectors $\beta \otimes \bigotimes_i \alpha_i$ span V_L^* . Their restrictions span E^* , forcing $L = E^*$, contrary to $d < r$. This proves the claim.

Starting from $J = \emptyset$, enlarge J using the claim until $d(J) = r$. Since each enlargement increases $d(J)$ by at least one, at most $r - 1$ factors are retained. A tuple attaining this rank gives the required covectors. $\square$

Apply the contraction lemma to both sides of a maximal-rank cut. The retained factors will serve as *anchors*: together they carry an invertible matrix that remains visible in each flattening separating the two anchor sets. Normalizing this matrix to the identity makes each remaining block equal to the product of its blocks in the distinguished row and column. The resulting identities give the following factorization.

Proposition 2.2 (Anchored matrix-product form). *Let $0 \neq T \in V_{[N]}$ have largest nontrivial flattening rank $r \geq 2$. There are disjoint nonempty sets $L, R \subseteq [N]$, each of cardinality at most $r - 1$, and, writing $C = [N] \setminus (L \cup R)$, there are subspaces $U \subseteq V_L$ and $W \subseteq V_R$ of dimension r , covectors $b_i \in V_i^*$ for $i \in C$, an identification $\Theta : \text{End}(\mathbb{C}^r) \rightarrow U \otimes W$, and linear maps*

$$M_i : V_i^* \longrightarrow \text{End}(\mathbb{C}^r) \quad (i \in C)$$

with the following properties. The tensor lies in $U \otimes W \otimes V_C$, and its contraction on the factors in C is

$$\Phi(x_C) := \Theta^{-1} \left(\left(\text{id}_{U \otimes W} \otimes \bigotimes_{i \in C} x_i \right) T \right) = \prod_{i \in C} M_i(x_i). \quad (6)$$

The empty product is the identity, each $M_i(b_i) = I_r$, and

$$[M_i(x), M_j(y)] = 0 \quad (i \neq j, x \in V_i^*, y \in V_j^*). \quad (7)$$

Consequently the product in (6) may be taken in any order.

Proof. Choose a cut $A \mid B$ of rank r . Its two Schmidt support spaces $E_A \subseteq V_A$ and $E_B \subseteq V_B$ have dimension r , and $T \in E_A \otimes E_B$ has matrix rank r in these spaces. Apply Lemma 2.1 separately to E_A and E_B . It yields retained sets $L \subseteq A$ and $R \subseteq B$, each of size at most $r - 1$, and product contractions injective on both support spaces. The tensor $T_0 \in V_L \otimes V_R$ obtained by applying both contractions therefore has rank r . Since $r \geq 2$, neither retained set is empty.

Contraction cannot increase flattening rank. It follows from the rank of T_0 that the original cuts $L \mid L^c$ and $R \mid R^c$ both have rank at least r , and hence exactly r . Let $U \subseteq V_L$ and $W \subseteq V_R$ be their respective support spaces on the retained sides. These have dimension r . Applying the two support projections shows that $T \in U \otimes W \otimes V_C$. In particular, $T_0 \in U \otimes W$ is a full-rank matrix.

Choose bases $u_1, \dots, u_r$ of U and $w_1, \dots, w_r$ of W such that $T_0 = \sum_{a=1}^r u_a \otimes w_a$, and set

$$\Theta(Z) := \sum_{a,b=1}^r Z_{ab} u_a \otimes w_b.$$

Denote the contracting covectors on C by b_i . The resulting matrix-valued multilinear map satisfies $\Phi(b_C) = I_r$. Define $M_i(x) = \Phi(b_{C \setminus \{i\}}, x)$; these maps are linear and $M_i(b_i) = I_r$.

Fix $S \subseteq C$. Extend each nonzero b_i to a basis of V_i^* . Arrange the coefficients of T across $(L \cup S) \mid (R \cup (C \setminus S))$ in a block matrix H , whose blocks are $r \times r$ matrices. The block rows are

indexed by basis tuples on S , and the block columns by basis tuples on $C \setminus S$. Label the all- b_i tuple by 0 on each side. Then $H_{0,0} = I_r$, and $\text{rank } H \leq r$ by the all-cut bound. Indeed, replacing the ambient spaces V_L, V_R by their supports U, W preserves the rank of this flattening.

Partition H around its distinguished block as

$$H = \begin{pmatrix} I_r & B_0 \\ C_0 & D_0 \end{pmatrix}.$$

Multiplication on the left and right by invertible block matrices gives

$$\begin{pmatrix} I_r & 0 \\ -C_0 & I \end{pmatrix} H \begin{pmatrix} I_r & -B_0 \\ 0 & I \end{pmatrix} = \begin{pmatrix} I_r & 0 \\ 0 & D_0 - C_0 B_0 \end{pmatrix}.$$

Thus $D_0 = C_0 B_0$, or equivalently $H_{\alpha,\beta} = H_{\alpha,0} H_{0,\beta}$ for all block indices. If one indexing set is empty, the same identity holds directly. By multilinearity, the block identities give

$$\Phi(x_S, x_{C \setminus S}) = \Phi(x_S, b_{C \setminus S}) \Phi(b_S, x_{C \setminus S}). \quad (8)$$

Extracting one singleton $S = \{i\}$ at a time proves (6). The singletons can be extracted in any order. In particular, keeping only x_i, x_j free and extracting them in the two possible orders gives $M_i(x_i) M_j(x_j) = M_j(x_j) M_i(x_i)$, proving (7). $\square$

The distinction between commutation across factors and commutation within a factor matters here. The cut identities never compare two independent choices of the variable at the same factor, so they do not force its coefficient algebra to be commutative. The next lemma shows that this is only a bounded obstruction. Each noncommutative algebra removes at least one dimension from the common centralizer, while all algebras belonging to other factors lie in that centralizer.

Lemma 2.3 (Centralizer bound). *Let $\mathcal{A}_i \subseteq \text{End}(\mathbb{C}^r)$ be finitely many unital subalgebras, and suppose that elements of $\mathcal{A}_i$ commute with elements of $\mathcal{A}_j$ whenever $i \neq j$. At most $r^2 - 1$ of the algebras $\mathcal{A}_i$ are noncommutative. All the remaining algebras together generate a commutative unital algebra of dimension at most r^2 .*

Proof. List the noncommutative algebras as $\mathcal{A}_{q_1}, \dots, \mathcal{A}_{q_b}$, and define

$$Z_0 = \text{End}(\mathbb{C}^r), \quad Z_k = \bigcap_{j=1}^k \text{Comm}(\mathcal{A}_{q_j}),$$

where $\text{Comm}(\mathcal{B})$ denotes the vector space of matrices commuting with every element of $\mathcal{B}$. Cross-site commutation implies $\mathcal{A}_{q_k} \subseteq Z_{k-1}$. Since $\mathcal{A}_{q_k}$ is noncommutative, there are $x, y \in \mathcal{A}_{q_k}$ with $[x, y] \neq 0$. Then $x \in Z_{k-1} \setminus Z_k$, so $Z_0 \supsetneq Z_1 \supsetneq \dots \supsetneq Z_b$. Every Z_k contains the scalar matrices. The chain therefore has at most $r^2 - 1$ strict inclusions. Finally, elements of the remaining algebras commute both within each algebra and across distinct algebras. They generate a commutative subalgebra of $\text{End}(\mathbb{C}^r)$, whose dimension is at most r^2 . $\square$

3 Border rank of products in commutative algebras

The preceding reduction controls the dimension of the algebra containing the remaining matrices, but leaves the number of factors unrestricted. Expanding their product term by term would reintroduce that dependence. The estimate in this section avoids such an expansion: in a local algebra of

dimension d , every product of d elements of the maximal ideal vanishes. The surviving terms are indexed by subsets of fewer than d factors, and polarization combines all subsets of a given size into coefficients of product curves. Finite differences bound the border rank of those coefficients using only d .

We will use two elementary border-rank properties. Local linear maps cannot increase border rank, as noted above. Also, $\underline{R}(S+T) \leq \underline{R}(S) + \underline{R}(T)$: choose convergent sequences of bounded-rank approximants to S and T and add the sequences. Tensoring with a fixed decomposable tensor cannot increase border rank, by tensoring the approximants with that tensor.

Lemma 3.1. *The two expressions in (2) agree. The function F is increasing on the positive integers and satisfies*

$$F(a) + F(b) \leq F(a+b) \quad (a, b \geq 1). \quad (9)$$

Proof. For $q \geq 2$, split the defining sum into its unweighted part and its part weighted by k . The identities

$$\sum_{k=0}^{q-1} \binom{q+k-2}{k} = \binom{2q-2}{q-1}, \quad k \binom{q+k-2}{k} = (q-1) \binom{q+k-2}{k-1} \quad (k \geq 1)$$

and summation of the second identity give the claimed closed form. Equivalently,

$$F(q) = \frac{q^2 - q + 1}{q} \binom{2q-2}{q-1},$$

which also holds at $q = 1$. Consequently, for $q \geq 2$,

$$\frac{F(q)}{F(q-1)} = \frac{2(2q-3)(q^2 - q + 1)}{q(q^2 - 3q + 3)} \geq 2.$$

The last inequality follows because the difference between $(2q-3)(q^2 - q + 1)$ and $q(q^2 - 3q + 3)$ is $q^2(q-2) + 2q - 3 > 0$. Thus F is increasing, and

$$F(a) + F(b) \leq 2F(a+b-1) \leq F(a+b),$$

as required. □

Proposition 3.2 (Border rank of an algebra product). *Let $\mathcal{A}$ be a commutative unital complex algebra of dimension $m \geq 1$. Let $n \geq 1$, let $\phi_i : V_i^* \rightarrow \mathcal{A}$ be linear maps, and let $\lambda \in \mathcal{A}^*$. The multilinear form*

$$H(x_1, \dots, x_n) = \lambda \left(\prod_{i=1}^n \phi_i(x_i) \right) \quad (10)$$

defines a tensor in $V_1 \otimes \dots \otimes V_n$ of border rank at most $F(m)$, independently of n and the local dimensions.

Proof. Decompose the finite-dimensional commutative algebra into its local Artin factors,

$$\mathcal{A} = \bigoplus_{\nu=1}^s \mathcal{A}_\nu, \quad d_\nu := \dim \mathcal{A}_\nu, \quad \sum_{\nu=1}^s d_\nu = m.$$

This decomposition follows from the Chinese remainder theorem applied to sufficiently high powers of the finitely many maximal ideals. Each residue field is $\mathbb{C}$, since it is a finite-dimensional field extension of $\mathbb{C}$. If $\mathfrak{m}_\nu$ is the maximal ideal of $\mathcal{A}_\nu$, then

$$\mathcal{A}_\nu = \mathbb{C}1_\nu \oplus \mathfrak{m}_\nu, \quad \dim \mathfrak{m}_\nu = d_\nu - 1, \quad \mathfrak{m}_\nu^{d_\nu} = 0.$$

For the last assertion, if two consecutive nonzero powers of $\mathfrak{m}_\nu$ were equal, Nakayama's lemma applied to that ideal as an $\mathcal{A}_\nu$ -module would force it to be zero. Hence their dimensions strictly decrease until zero; starting at dimension $d_\nu - 1$ gives the asserted exponent.

Let $\phi_{i,\nu}$ and λ_ν denote the components of ϕ_i and λ on $\mathcal{A}_\nu$. Multiplication is componentwise, so $H = \sum_\nu H_\nu$, where $H_\nu(x) = \lambda_\nu(\prod_i \phi_{i,\nu}(x_i))$. Write

$$\phi_{i,\nu}(x_i) = \alpha_{i,\nu}(x_i)1_\nu + n_{i,\nu}(x_i),$$

with $\alpha_{i,\nu} : V_i^* \rightarrow \mathbb{C}$ and $n_{i,\nu} : V_i^* \rightarrow \mathfrak{m}_\nu$ linear. Fix ν and write $d = d_\nu$. Expanding the product and using $\mathfrak{m}_\nu^d = 0$ gives

$$H_\nu(x) = \sum_{k=0}^{d-1} \sum_{\substack{S \subseteq [n] \\ |S|=k}} \lambda_\nu \left(\prod_{i \in S} n_{i,\nu}(x_i) \right) \prod_{j \notin S} \alpha_{j,\nu}(x_j). \quad (11)$$

The empty algebra product in the $k = 0$ term is 1_ν , so this term has rank at most one. For $k > 0$, bounding each subset separately would cost $\binom{n}{k}$ summands. We instead treat the symmetric multiplication form once and use the same decomposition for every subset.

For $1 \leq k \leq d-1$, commutativity makes

$$(y_1, \dots, y_k) \mapsto \lambda_\nu(y_1 \cdots y_k)$$

a symmetric k -linear form on $\mathfrak{m}_\nu$. Pure powers $\eta^{\otimes k}$, with $\eta \in \mathfrak{m}_\nu^*$, span $\text{Sym}^k(\mathfrak{m}_\nu^*)$. Indeed, a linear functional annihilating all such powers gives a homogeneous polynomial vanishing at every η , and hence is zero; equivalently this is polarization in characteristic zero. Selecting a basis from these powers therefore expresses the displayed form as

$$\lambda_\nu(y_1 \cdots y_k) = \sum_{\ell=1}^{h_{d,k}} c_\ell \eta_\ell(y_1) \cdots \eta_\ell(y_k), \quad h_{d,k} \leq \binom{d+k-2}{k}. \quad (12)$$

The linear forms η_ℓ and coefficients c_ℓ depend only on the multiplication form at degree k , and hence apply to every subset in (11). Zero coefficients or fewer summands are allowed.

Fix one summand and put $\beta_i = \eta_\ell \circ n_{i,\nu}$. We identify the linear forms $\alpha_{i,\nu}, \beta_i$ on V_i^* with vectors in V_i . The corresponding tensor in the degree- k part of (11) is the coefficient of t^k in

$$P(t) := \bigotimes_{i=1}^n (\alpha_{i,\nu} + t\beta_i).$$

For $k \geq 1$, it has the explicit degeneration

$$[t^k]P(t) = \frac{1}{k!} \lim_{\varepsilon \rightarrow 0} \varepsilon^{-k} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} P(j\varepsilon). \quad (13)$$

To verify this, expand $P(t) = \sum_h P_h t^h$. The k -th forward difference of a polynomial of degree less than k is zero, while that of t^k is the constant $k!$. Thus the coefficients with $h < k$ cancel, the coefficient with $h = k$ becomes P_k , and every coefficient with $h > k$ is multiplied by a positive power of ε and tends to zero. For every nonzero ε , the sum on the right of (13) has CP rank at most $k+1$, since each $P(j\varepsilon)$ is decomposable or zero. Therefore $\underline{\text{R}}([t^k]P(t)) \leq k+1$.

Subadditivity, (12), and the rank-one $k = 0$ term imply, for $d \geq 2$,

$$\underline{\text{R}}(H_\nu) \leq 1 + \sum_{k=1}^{d-1} (k+1) \binom{d+k-2}{k} = F(d).$$

For $d = 1$, only the rank-one term is present and the same conclusion holds with $F(1) = 1$. Finally,

$$\underline{R}(H) \leq \sum_{\nu=1}^s F(d_\nu) \leq F\left(\sum_{\nu=1}^s d_\nu\right) = F(m)$$

by Lemma 3.1. □

The key uniformity is visible in (13): the number of evaluation points depends on the coefficient degree k , which is bounded by the local algebra dimension, and is independent of n . The degeneration occurs in the tensor space. Thus the argument also applies to algebras that are not limits of reduced algebras.

4 Proof of the uniform bound

Only the anchor factors and the noncommutative factors need to be expanded in local bases. Their number is bounded by $r^2 + 2r - 3$; all other factors remain together in a single algebra product. The expansion must use bases of the original local spaces, since a vector in an anchor support $U \subseteq V_L$ need not be decomposable across L . This accounts for the power of r in the theorem, while the algebra product contributes $F(r^2)$.

Proof of Theorem 1.1. Reduce each V_i to its local support, as at the start of Section 2. This preserves border rank and all flattening ranks and gives $\dim V_i \leq r$. If $r = 1$, every local support is one-dimensional, so T is decomposable. Assume henceforth that $r \geq 2$.

Apply Proposition 2.2, with anchor sets L, R , remaining set C , support spaces U, W , identification Θ , and matrix maps M_i . For each $i \in C$, let $\mathcal{A}_i$ be the unital algebra generated by $M_i(V_i^*)$. The cross-site commutation relations extend to these generated algebras, because sums and products of matrices commuting with a fixed matrix continue to commute with it. Let $Q \subseteq C$ be the set of indices for which $\mathcal{A}_i$ is noncommutative, and put $G = C \setminus Q$. By Lemma 2.3,

$$|Q| \leq r^2 - 1. \quad (14)$$

The algebras $\mathcal{A}_i$, $i \in G$, generate a commutative unital algebra $\mathcal{A} \subseteq \text{End}(\mathbb{C}^r)$, with $\dim \mathcal{A} \leq r^2$. If $G = \emptyset$, take $\mathcal{A} = \mathbb{C}I_r$.

Set $K = L \cup R \cup Q$. Then

$$|K| \leq 2(r - 1) + (r^2 - 1) = r^2 + 2r - 3. \quad (15)$$

For each $i \in K$, choose a basis $(e_{i,a})_{a=1}^{d_i}$ of V_i , where $d_i \leq r$, with dual basis $(e_{i,a}^*)_{a=1}^{d_i}$. Expanding only on these factors gives

$$T = \sum_{\gamma \in \prod_{i \in K} [d_i]} \left(\bigotimes_{i \in K} e_{i,\gamma_i} \right) \otimes T_\gamma, \quad T_\gamma \in V_G. \quad (16)$$

There are $\prod_{i \in K} d_i \leq r^{|K|}$ summands, and every displayed factor on K is decomposable in the original tensor factors.

Suppose first that $G \neq \emptyset$. For a fixed γ , define

$$B_\gamma := \prod_{i \in Q} M_i(e_{i,\gamma_i}^*), \quad u_\gamma^* := \bigotimes_{i \in L} e_{i,\gamma_i}^* \Big|_U, \quad w_\gamma^* := \bigotimes_{i \in R} e_{i,\gamma_i}^* \Big|_W.$$

The empty product for B_γ is I_r . These matrices may be multiplied in any order, since they come from distinct sites. Define a linear functional on $\mathcal{A}$ by

$$\lambda_\gamma(a) := (u_\gamma^* \otimes w_\gamma^*)(\Theta(B_\gamma a)).$$

The anchored formula, with the factors in Q evaluated at their chosen dual basis vectors, gives

$$T_\gamma(x_G) = \lambda_\gamma \left(\prod_{i \in G} M_i(x_i) \right). \quad (17)$$

Here B_γ need not belong to $\mathcal{A}$; it only enters the linear functional λ_γ . Each $M_i(x_i)$, $i \in G$, does belong to $\mathcal{A}$. Proposition 3.2 and monotonicity of F therefore give

$$\underline{R}(T_\gamma) \leq F(\dim \mathcal{A}) \leq F(r^2).$$

When $G = \emptyset$, the coefficients in (16) are scalars, so each summand in that expansion already has rank at most one.

In either case, subadditivity and (16) yield

$$\underline{R}(T) \leq r^{|K|} F(r^2) \leq r^{r^2+2r-3} F(r^2),$$

which proves the theorem. $\square$

The argument also covers one-dimensional local factors. A nonzero order-one tensor has rank and border rank one; it is excluded from the statement only because there is then no nontrivial cut over which to define r .

5 Cactus varieties and algebraic branching programs

5.1 Cactus varieties

The algebra-product estimate has a geometric consequence independent of the anchored reduction. For a projective variety X , let $\kappa_m(X)$ denote the Zariski closure of the union of the linear spans of finite subschemes of X of length at most m . This is the m -th cactus variety, with the stated closure convention.

Corollary 5.1 (Cactus-to-secant containment for Segre varieties). *Let $X = \text{Seg}(\mathbb{P}V_1 \times \cdots \times \mathbb{P}V_n)$, where $n \geq 1$ and the V_i are nonzero finite-dimensional complex vector spaces. For every integer $m \geq 1$,*

$$\kappa_m(X) \subseteq \sigma_{F(m)}(X).$$

The bound is independent of the number and dimensions of the factors.

Proof. Let $Z \subseteq X$ be a nonempty finite subscheme of length $1 \leq d \leq m$. In each factor choose a homogeneous linear form $\ell_i \in V_i^*$ that is nonzero at every point in the projection of the finite support of Z . This is possible because only finitely many proper hyperplanes in V_i^* are excluded. The resulting product affine chart contains Z , and the section $\ell_1 \cdots \ell_n$ trivializes $\mathcal{O}_X(1, \dots, 1)|_Z$.

The coordinate algebra $\mathcal{A} = H^0(Z, \mathcal{O}_Z)$ has dimension d . Restriction in this trivialization gives linear maps

$$\phi_i : V_i^* \longrightarrow \mathcal{A}, \quad x_i \longmapsto (x_i/\ell_i)|_Z.$$

The restriction of a decomposable global section $x_1 \cdots x_n$ is $\prod_i \phi_i(x_i)$. The affine cone over the linear span of Z is the image of the dual of this restriction map: its annihilator consists exactly of the linear forms vanishing on Z . Consequently, every tensor in that cone has the form $\lambda(\prod_i \phi_i(x_i))$ for some $\lambda \in \mathcal{A}^*$. Proposition 3.2 gives border rank at most $F(d) \leq F(m)$. Taking the union over Z and then its Zariski closure proves the assertion, since $\sigma_{F(m)}(X)$ is closed. $\square$

This containment has a larger secant index and does not identify cactus rank with border rank. The distinction is substantive: Doležálek and Michalek construct equations separating the m -th secant from the m -th cactus variety for three-factor Segre varieties with $m \geq 14$ [10, Theorem 1.1]. Corollary 5.1 is compatible with such separations because it permits the loss from m to $F(m)$.

5.2 Examples and algebraic branching programs

The finite-difference construction also explains the fixed-weight examples directly. Given vectors $e_{i,0}, e_{i,1} \in V_i$, put

$$D_{N,k} := [t^k] \bigotimes_{i=1}^N (e_{i,0} + te_{i,1}), \quad 0 \leq k \leq N.$$

Equation (13) gives $\underline{R}(D_{N,k}) \leq k + 1$, with the case $k = 0$ being decomposable. Across any cut, sorting the chosen factors by the number on each side writes $D_{N,k}$ as a sum of at most $k + 1$ bipartite product terms, so its flattening ranks are at most $k + 1$ as well. For independent pairs $e_{i,0}, e_{i,1}$, the case $k = 1$ is the unnormalized W tensor. Its exact rank grows with N , whereas its border rank is two for $N \geq 2$, illustrating why the main theorem cannot be strengthened by simply replacing border rank with rank [1, Section VIII].

There is a parallel formulation in algebraic complexity. In local bases, associate to T the set-multilinear polynomial

$$f_T(x_1, \dots, x_N) = \sum_{a_1, \dots, a_N} T_{a_1, \dots, a_N} x_{1,a_1} \cdots x_{N,a_N},$$

where $x_i = (x_{i,1}, \dots, x_{i, \dim V_i})$ is the i -th variable group. A matrix-product representation with one homogeneous linear matrix per group is a set-multilinear read-once oblivious algebraic branching program. Its minimum width at each layer is the corresponding prefix flattening rank, by the same successive-factorization argument as for MPS. The all-cut hypothesis is consequently a width bound in every group order. Any-order branching programs and identity testing are studied in [7]; the distinction between cross-layer commutation, full commutation, and diagonal representations, and their connections with tensor and Waring rank, is developed in [8].

Bhargava and Tengse construct a commutative set-multilinear branching program of width at most the dimension of the full partial-derivative space [9, Theorem 1.6]. This parameter differs from the largest flattening rank. If $\mathcal{D}(f_T)$ denotes that full space, including the zeroth derivative, then

$$\dim \mathcal{D}(f_T) = \sum_{S \subseteq [N]} \text{rank Flat}_{S|S^c}(T) \leq 2^N r. \quad (18)$$

Indeed, derivatives taken once in each group indexed by S span the image of that flattening, and distinct subsets give distinct remaining multidegrees. Even for a nonzero decomposable tensor, every summand in (18) has dimension one, so the full dimension is 2^N . Thus the full derivative dimension does not itself yield an order-independent bound from the all-cut hypothesis. The exponential loss in our theorem also does not resolve the polynomial-simulation questions for the structured branching-program classes in [8].

The all-cut hypothesis also limits the number of independent entangled blocks. Suppose a tensor is a product over disjoint blocks of factors, and s of those blocks each have an internal cut of rank at least two. Choose one such cut in every block and combine them into a cut of the whole tensor, placing any remaining blocks entirely on one side. The resulting flattening is a tensor product of matrices, so its rank is at least 2^s . Hence a tensor with maximal cut rank r can have at most $\lfloor \log_2 r \rfloor$ such blocks. This observation does not by itself control tensors without a block-product decomposition; the anchored factorization supplies the needed replacement.

The estimate $F(q) \leq q^{4^{q-1}}$ gives $\hat{B}(D) = \exp(O(D^2 \log D))$. The optimal growth of $B_*(D)$, including its value at $D = 3$, remains undetermined. The proof also identifies the difficulty in obtaining a stable version from approximately low-rank flattenings: the invertible core may be ill-conditioned, and the finite-difference coefficients can diverge. Controlling these two effects would be necessary for a quantitative approximation guarantee. All statements here concern complex border rank; the local decomposition used in the proof takes its residue fields to be $\mathbb{C}$.

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